Homework 5

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5-2. 推迟的场叶菲缅科 (Jefimenko) 公式. (1)

Proof. 真空中的麦克斯韦方程

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0, \tag{1.1}$$

$$\nabla \times \mathbf{B} - \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}, \tag{1.2}$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \tag{1.3}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{1.4}$$

对 (1.3) 式取旋度得到

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} + \frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = 0,$$

再代入 (1.1) 式和 (1.2) 式，

$$\frac{1}{\epsilon_0}\nabla \rho - \nabla^2 \mathbf{E} + \frac{\partial}{\partial t}\left[\frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}\right] = 0$$

$$\Rightarrow \left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right]\mathbf{E}(\mathbf{x}, t) = -\frac{1}{\epsilon_0}\left[-\nabla \rho - \frac{1}{c^2}\frac{\partial \mathbf{J}}{\partial t}\right]. \tag{1.5}$$

同样地，对 (1.2) 式取旋度并结合 (1.4) 得到

$$-\nabla^2 \mathbf{B} - \frac{1}{c^2}\frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = \mu_0 \nabla \times \mathbf{J},$$

代入 (1.3) 式，

$$\left[\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right]\mathbf{B}(\mathbf{x}, t) = -\mu_0 \nabla \times \mathbf{J}. \tag{1.6}$$

□

(2) 利用 Fourier 变换可以解出符合物理因果律的推迟格林函数

$$G^{(+)}(R,\tau) = \frac{\delta\left(t' - \left[t - \frac{R}{c}\right]\right)}{R},$$

由此可以积分得到波动方程的解为

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\int \mathrm{d}^3\mathbf{x}'\,\mathrm{d}t'\,\frac{\delta\left(t' - \left[t - \frac{R}{c}\right]\right)}{R}\left[-\nabla'\rho - \frac{1}{c^2}\frac{\partial \mathbf{J}}{\partial t'}\right] = \frac{1}{4\pi\epsilon_0}\int \mathrm{d}^3\mathbf{x}'\,\frac{1}{R}\left[-\boldsymbol{\nabla}\rho - \frac{1}{c^2}\frac{\partial \mathbf{J}}{\partial t}\right]_{\text{ret}}, \tag{1.7}$$

$$\mathbf{B}(\mathbf{x},t) = \frac{\mu_0}{4\pi}\int \mathrm{d}^3\mathbf{x}'\,\mathrm{d}t'\,\frac{\delta\left(t' - \left[t - \frac{R}{c}\right]\right)}{R}[\nabla'\times\mathbf{J}] = \frac{\mu_0}{4\pi}\int \mathrm{d}^3\mathbf{x}'\,\frac{1}{R}[\nabla'\times\mathbf{J}]_{\text{ret}}. \tag{1.8}$$

4. 推导长波近似下的电偶极辐射电磁场. 书 (5.19) 式给出了长波近似下的磁矢势

$$\mathbf{A}(\mathbf{x}) = -\frac{\mathrm{i}\mu_0\omega}{4\pi}\mathbf{p}\frac{\mathrm{e}^{\mathrm{i}kr}}{r}. \tag{5.19}$$

利用书 (5.14) 式以及书 (5.20) 式给出的矢量分析公式，有磁场

$$\begin{aligned}\mathbf{H} = \frac{1}{\mu_0}\boldsymbol{\nabla}\times\mathbf{A} &= -\frac{\mathrm{i}\omega}{4\pi}\boldsymbol{\nabla}\times\left(\mathbf{p}\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\right) = -\frac{\mathrm{i}\omega}{4\pi}\boldsymbol{\nabla}\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\times\mathbf{p}\\ &= \frac{ck^2}{4\pi}(\mathbf{n}\times\mathbf{p})\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\left(1 - \frac{1}{\mathrm{i}kr}\right).\end{aligned} \tag{1.9}$$

进而有

$$\begin{aligned}\mathbf{E} = \frac{\mathrm{i}Z_0}{k}\boldsymbol{\nabla}\times\mathbf{H} &= \frac{\mathrm{i}k}{4\pi\epsilon_0}\boldsymbol{\nabla}\times\left[(\mathbf{n}\times\mathbf{p})\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\left(1 - \frac{1}{\mathrm{i}kr}\right)\right]\\ &= \frac{\mathrm{i}k}{4\pi\epsilon_0}\left\{\boldsymbol{\nabla}\left[\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\left(1 - \frac{1}{\mathrm{i}kr}\right)\right]\times(\mathbf{n}\times\mathbf{p}) + \frac{\mathrm{e}^{\mathrm{i}kr}}{r}\left(1 - \frac{1}{\mathrm{i}kr}\right)\boldsymbol{\nabla}\times(\mathbf{n}\times\mathbf{p})\right\}\\ &= \frac{\mathrm{i}k}{4\pi\epsilon_0}\left\{\left(\mathrm{i}k - \frac{2}{r} + \frac{2}{\mathrm{i}kr^2}\right)\frac{\mathrm{e}^{\mathrm{i}kr}}{r}[\mathbf{n}\times(\mathbf{n}\times\mathbf{p})] + \frac{1}{\mathrm{i}k}\left(\frac{1}{r^3} - \frac{\mathrm{i}k}{r^2}\right)[\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}]\right\}\\ &= \frac{1}{4\pi\epsilon_0}\left\{k^2(\mathbf{n}\times\mathbf{p})\times\mathbf{n}\frac{\mathrm{e}^{\mathrm{i}kr}}{r} + [3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}]\left(\frac{1}{r^3} - \frac{\mathrm{i}k}{r^2}\right)\mathrm{e}^{\mathrm{i}kr}\right\}.\end{aligned} \tag{1.10}$$

郭书 5.10 在直角坐标中可以将磁偶极矩表示为

$$\mathbf{m} = \frac{4\pi}{3}M_0R_0^3\mathrm{e}^{-\mathrm{i}\omega t}(\mathbf{e}_x + \mathrm{i}\mathbf{e}_y),$$

转换到球坐标中，

$$\mathbf{m} = \frac{4\pi}{3}M_0R_0^3\mathrm{e}^{-\mathrm{i}\omega t + \mathrm{i}\phi}(\sin\theta\mathbf{e}_R + \cos\theta\mathbf{e}_\theta + \mathrm{i}\mathbf{e}_\phi).$$

带入辐射电磁场磁偶极项表达式

$$
\begin{aligned}
\mathbf{B} &= \frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi c^2 R}(\ddot{\mathbf{m}} \times \mathbf{e}_R) \times \mathbf{e}_R = \frac{\mu_0 \omega^2 R_0^3 M_0}{3c^2 R}(\mathbf{e}_\theta \cos\theta + \mathrm{i}\mathbf{e}_\phi)\mathrm{e}^{\mathrm{i}(kR-\omega t+\phi)}, \\
\mathbf{E} &= -\frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi c R}(\ddot{\mathbf{m}} \times \mathbf{e}_R) = \frac{\mu_0 \omega^2 R_0^3 M_0}{3cR}(\mathrm{i}\mathbf{e}_\theta - \mathbf{e}_\phi \cos\theta)\mathrm{e}^{\mathrm{i}(kR-\omega t+\phi)},
\end{aligned}
$$

故有

$$
\mathbf{S} = \frac{1}{2\mu_0}\operatorname{Re}(\mathbf{E}^* \times \mathbf{B}) = \frac{\mu_0 \omega^4 R_0^6 M_0^2}{18c^3 R^2}(1+\cos^2\theta)\mathbf{e}_R.
$$

郭书 5.11 直角坐标中电偶极矩表示为

$$
\mathbf{p} = ea\mathrm{e}^{-\mathrm{i}\omega t}(\mathbf{e}_x + \mathrm{i}\mathbf{e}_y),
$$

相应的球坐标表示为

$$
\mathbf{p} = ea\mathrm{e}^{-\mathrm{i}\omega t+\mathrm{i}\phi}(\sin\theta\mathbf{e}_R + \cos\theta\mathbf{e}_\theta + \mathrm{i}\mathbf{e}_\phi).
$$

带入辐射电磁场电偶极项表达式

$$
\begin{aligned}
\mathbf{E} &= \frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi\epsilon_0 c^2 R}(\ddot{\mathbf{p}} \times \mathbf{e}_R) \times \mathbf{e}_R = \frac{\omega^2 ea}{4\pi\epsilon_0 c^2 R}(\mathbf{e}_\theta \cos\theta + \mathrm{i}\mathbf{e}_\phi)\mathrm{e}^{\mathrm{i}(kR-\omega t+\phi)}, \\
\mathbf{B} &= \frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi\epsilon_0 c^3 R}\ddot{\mathbf{p}} \times \mathbf{e}_R = \frac{\omega^2 ea}{4\pi\epsilon_0 c^3 R}(\mathbf{e}_\phi \cos\theta - \mathrm{i}\mathbf{e}_\theta)\mathrm{e}^{\mathrm{i}(kR-\omega t+\phi)},
\end{aligned}
$$

辐射能流

$$
\mathbf{S} = \frac{1}{2\mu_0}\operatorname{Re}(\mathbf{E}^* \times \mathbf{B}) = \frac{\mu_0 \omega^4 e^2 a^2}{32\pi^2 c R^2}(1+\cos^2\theta)\mathbf{e}_R.
$$

8-1.